

H¹-in- ρ Bound for the Centroid of the Torsion Level Family (Route δ , Plan 2, Building Block CENTROIDBOUND)

Abstract

This note is the building block CENTROIDBOUND of the Plan 2 proof of sharp quantitative Faber–Krahn stability via Route δ (no graph, no selection, no Schauder). For a near-spherical torsion domain $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and small Saint–Venant deficit $\delta := \delta_T(\Omega)$, let $E_\rho := \{u > t(\rho)\}$ be the volume-radius parametrised superlevels of the torsion function u and let

$$\bar{z}_\rho := \frac{1}{|E_\rho|} \int_{E_\rho} x \, dx \in \mathbb{R}^n$$

be the *centroid* of E_ρ . We prove that $\rho \mapsto \bar{z}_\rho$ is Lipschitz on $[\rho_*, 1]$ and satisfies the integrated H^1 -in- ρ bound

$$\int_{\rho_*}^{\rho_\delta} |\bar{z}'_\rho|^2 (-t'(\rho)) \, d\rho \leq C(n, R, \rho_*) \delta_T(\Omega).$$

The proof uses the centroid kinematic identity (a bulk-coarea computation that has *no* homothety term), the unconditional D_H -only Cauchy–Schwarz of the velocity defect, the Figalli–Maggi–Pratelli volume Fraenkel estimate, the Fusco–Julin L^1 oscillation, the divergence theorem for finite-perimeter sets, and Agent B’s integrated Talenti gap. No graph parametrisation, no selection principle, no pointwise quantitative isoperimetric form is invoked.

1 Statement and main result

Fix $n \geq 2$, $R \geq 2$, $\rho_* \in (0, 1)$, and $\kappa > 0$. Let $\Omega \subset B_R \subset \mathbb{R}^n$ be a bounded open set with $|\Omega| = \omega_n := |B_1|$ and Saint–Venant deficit $\delta := \delta_T(\Omega) \leq \delta_0(n, R, \rho_*)$. Let $u \in W_0^{1,2}(\Omega) \cap W_{\text{loc}}^{2,p}(\Omega)$ ($p > n$) be the variational torsion function of Ω , $t : [\rho_*, 1] \rightarrow (0, \|u\|_\infty)$ the volume-radius parametrisation defined by $|\{u > t(\rho)\}| = \omega_n \rho^n$, $E_\rho := \{u > t(\rho)\}$, and

$$V_\rho(x) := \frac{-t'(\rho)}{|\nabla u(x)|} \quad \text{on } \partial^* E_\rho \cap \{|\nabla u| > 0\}.$$

Set $\rho_\delta := 1 - \kappa\sqrt{\delta}$.

Definition 1.1 (Centroid field). For $\rho \in [\rho_*, 1]$ define the *centroid* of E_ρ by

$$\bar{z}_\rho := \frac{1}{|E_\rho|} \int_{E_\rho} x \, dx \in \mathbb{R}^n. \tag{1}$$

This is the bulk barycentre; it is *not* obtained by Fraenkel minimisation and inherits no implicit-function-theorem regularity.

Theorem 1.2 (H^1 -in- ρ centroid bound). *There exist $\delta_0 = \delta_0(n, R, \rho_*) > 0$ and $C_{\text{cent}} = C_{\text{cent}}(n, R, \rho_*) > 0$ such that for every bounded open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta \leq \delta_0$, the centroid field $\rho \mapsto \bar{z}_\rho$ is Lipschitz on $[\rho_*, 1]$ with explicit constant*

$$L_{\text{cent}} := \frac{2nR}{\rho_*^n}, \tag{2}$$

in particular absolutely continuous, and

$$\int_{\rho_*}^{\rho_\delta} |\bar{z}'_\rho|^2 (-t'(\rho)) d\rho \leq C_{\text{cent}} \delta_T(\Omega). \quad (\text{F})$$

Remark 1.3 (Compatibility with the Fraenkel centre). By Cicalese–Leonardi [3, Prop. 2.3] and Acerbi–Fusco–Morini [1, Lem. 2.6], in the smallness regime $|\bar{z}_\rho - z_\rho^{\text{Fr}}| \leq C \mathcal{A}(E_\rho) \leq C' \sqrt{\delta_T(\Omega)}$. The centroid is therefore interchangeable with z_ρ^{Fr} in downstream uses at cost $O(\sqrt{\delta_T})$, absorbed into constants exactly as in Wave 2 C §C1.

2 Setup, measurability, Lipschitz-in- ρ

2.1 The smallness regime

By the smallness propagation (Figalli–Maggi–Pratelli [5] applied levelwise plus the integrated $\int D_I d\mu \leq C\delta_T$ of [9]), $\mathcal{A}(E_\rho) \leq C(n, \rho_*) \sqrt{\delta_T(\Omega)} \leq \varepsilon_0$ for all $\rho \in [\rho_*, 1]$, with ε_0 the CL 2012 smallness threshold. The CL containment

$$B(z_\rho^{\text{Fr}}, \rho/2) \subset E_\rho \subset B(z_\rho^{\text{Fr}}, 2\rho) \quad (3)$$

holds for a.e. ρ . Combined with the centroid–Fraenkel proximity, $B(\bar{z}_\rho, \rho/4) \subset E_\rho \subset B(\bar{z}_\rho, 3\rho)$. In particular $|x - \bar{z}_\rho| \leq 3\rho \leq 3$ on $\partial^* E_\rho$ and $\bar{z}_\rho \in E_\rho^\circ$.

2.2 Borel measurability and Lipschitz

Lemma 2.1. $\rho \mapsto \bar{z}_\rho$ is Borel and Lipschitz on $[\rho_*, 1]$ with constant $L_{\text{cent}} = 2nR/\rho_*^n$.

Proof. Borel. Since t is decreasing, $\rho \mapsto E_\rho$ is monotone nested; dominated convergence gives $\rho \mapsto \int_{E_\rho} x_i dx$ continuous, hence Borel.

Lipschitz. For $\rho_* \leq \rho \leq \rho' \leq 1$, nesting gives

$$|E_\rho \Delta E_{\rho'}| = |E_{\rho'}| - |E_\rho| = \omega_n((\rho')^n - \rho^n) \leq n\omega_n|\rho - \rho'|.$$

Since $E_\rho, E_{\rho'} \subset B_R$, $|\int x_i - \int x_i| \leq Rn\omega_n|\rho - \rho'|$. Combined with $|E_\rho| \geq \omega_n\rho_*^n$, $|\bar{z}_\rho - \bar{z}_{\rho'}| \leq (2nR/\rho_*^n)|\rho - \rho'|$. $\square$

3 The centroid kinematic identity

Lemma 3.1 (Centroid kinematic formula). For a.e. $\rho \in [\rho_*, 1]$,

$$\bar{z}'_\rho = \frac{1}{|E_\rho|} \int_{\partial^* E_\rho} (x - \bar{z}_\rho) V_\rho d\mathcal{H}^{n-1}. \quad (4)$$

Proof. By the coarea identity [9, §3] and Sobolev–Sard [4, App. A, Thm. A.1], for every Lipschitz $f : B_R \rightarrow \mathbb{R}$ and a.e. $\rho \in [\rho_*, 1]$, $\frac{d}{d\rho} \int_{E_\rho} f dx = \int_{\partial^* E_\rho} f V_\rho d\mathcal{H}^{n-1}$. Taking $f(x) = x_i$ and applying the product rule:

$$\bar{z}'_{\rho,i} = \frac{1}{|E_\rho|} \int_{\partial^* E_\rho} x_i V_\rho dH - \frac{P(B_\rho)}{|E_\rho|^2} \int_{E_\rho} x_i dx = \frac{1}{|E_\rho|} \int_{\partial^* E_\rho} (x_i - \bar{z}_{\rho,i}) V_\rho dH,$$

using $\int V_\rho dH = P(B_\rho)$ (coarea with $f \equiv 1$). $\square$

Remark 3.2 (No homothety term — the structural mechanism). Identity (4) is the first vector moment of V_ρ relative to the centroid. *No homothetic vector field* $H_{z,\rho}(x) = (x - z) \cdot \nu_E / \rho$ enters. By contrast, defining the centre field via the Fraenkel functional first-order condition would yield a kinematic formula whose right-hand side involves $V_\rho - H_{z,\rho}$, and the bound would consume the per- ρ homothetic velocity defect of `gmt-inputs-for-metric-closure.md` (2.4), itself conditional on the per- ρ radial trace $(R)/(1.2)$, which is open in the no-graph regime. The centroid choice eliminates the homothety because it is defined directly by a bulk integral, not by a minimisation principle comparing to balls. This is precisely the structural mechanism by which Theorem 1.2 avoids (R) , (B^2) , the per- ρ pointwise radial-trace inputs of the Wave-1 audit (`gmt-inputs-for-metric-closure.md` (1.1), $(1.2)=(R)$, (2.4) in the source notation) entirely.

4 Pointwise bound $|\bar{z}'_\rho|^2 \leq C(D_H + D_I)$

We bound the integrand in (F) pointwise by the two algebraic deficits $D_H(t(\rho)), D_I(t(\rho))$ of [9].

Lemma 4.1 (Pointwise centroid-velocity bound). *There is $C_* = C_*(n, R, \rho_*)$ such that for a.e. $\rho \in [\rho_*, 1]$ in the smallness regime,*

$$|\bar{z}'_\rho|^2 \leq C_*(D_H(t(\rho)) + D_I(t(\rho))). \quad (5)$$

Proof. Set $\bar{V}_\rho := P(B_\rho)/P(E_\rho) \in (0, 1]$. Decompose $V_\rho = \bar{V}_\rho + (V_\rho - \bar{V}_\rho)$ in (4):

$$|E_\rho| \bar{z}'_\rho = \underbrace{\bar{V}_\rho \int (x - \bar{z}_\rho) d\mathcal{H}^{n-1}}_{(I)} + \underbrace{\int (x - \bar{z}_\rho)(V_\rho - \bar{V}_\rho) d\mathcal{H}^{n-1}}_{(II)}. \quad (6)$$

Bound on (II). By CL, $|x - \bar{z}_\rho| \leq 3$, so $|(II)| \leq 3 \int |V_\rho - \bar{V}_\rho|$. By the unconditional Cauchy–Schwarz on V_ρ , an algebraic consequence of [9]: writing $\bar{V}_\rho = A_\rho/P(E_\rho)$ and applying $\int |f| \leq (\int f^2/V)^{1/2} (\int V)^{1/2}$ with $f = |V_\rho - \bar{V}_\rho|$, $V = V_\rho$, together with the exact identity $\int (V_\rho - \bar{V}_\rho)^2/V_\rho d\mathcal{H}^{n-1} = (A_\rho/P(E_\rho)^2)D_H$,

$$\left(\int |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C_{n,\rho_*} D_H(t(\rho)). \quad (7)$$

Bound on (I). We claim, for each i ,

$$\left| \int_{\partial^* E_\rho} (x - \bar{z}_\rho)_i d\mathcal{H}^{n-1} \right| \leq C_2(n, \rho_*) \sqrt{\delta_\rho}, \quad (8)$$

with $\delta_\rho := P(E_\rho) - P(B_\rho)$. To see this, write $(x - \bar{z}_\rho)_i = |x - \bar{z}_\rho|(e_r)_i$, and split $(e_r)_i = (\nu_E)_i + ((e_r)_i - (\nu_E)_i)$.

For the first piece, apply the divergence theorem to the Lipschitz field $V^{(i)}(x) := |x - \bar{z}_\rho| e_i$ ($W^{1,\infty}$, kink at $\bar{z}_\rho$ is $\mathcal{H}^{n-1}$ -null): $\partial_j V_j^{(i)} = (e_r)_i$, so by [7, Thm. 15.9],

$$\int_{\partial^* E_\rho} |x - \bar{z}_\rho| (\nu_E)_i d\mathcal{H}^{n-1} = \int_{E_\rho} (e_r(x))_i dx. \quad (9)$$

On $B_\rho(\bar{z}_\rho)$, $\int (e_r)_i = 0$ by spherical symmetry. Hence $|\int_{E_\rho} (e_r)_i| \leq |E_\rho \Delta B_\rho(\bar{z}_\rho)|$. By FMP [5] and the centroid-Fraenkel proximity, $|E_\rho \Delta B_\rho(\bar{z}_\rho)| \leq C \sqrt{\delta_\rho}$.

For the second piece, $|x - \bar{z}_\rho| \leq 3$ and the FJ L^1 oscillation [6] at the centroid (with $O(\sqrt{\delta_\rho})$ change-of-centre error absorbed): $\int |\nu_E - e_r| \leq C_{FJ} \sqrt{\delta_\rho}$.

Combining gives (8).

Combining. From (6), (7), (8):

$$|E_\rho|^2 |\bar{z}'_\rho|^2 \leq 2C_2^2 \delta_\rho + 2C_1 D_H,$$

where $C_1 = 9C_{n,\rho_*}$. Use $\delta_\rho \leq D_I/(n\omega_n \rho_*^{n-1})$ and $|E_\rho|^2 \geq \omega_n^2 \rho_*^{2n}$ to get (5). $\square$

5 Closing the integral

Proof of Theorem 1.2. By Lemma 4.1 and the algebraic identity $D_H + D_I = P(B_\rho)^2|\psi|/(-t')$ (immediate from [9]) ($D_H + D_I = P(B_\rho)^2|\psi|/(-t')$ with $\psi := -t'_B - (-t')$), and the integrated Talenti gap $\int_{\rho_*}^1 |\psi(\rho)| d\rho \leq K_1\delta$, $K_1 = 2/(\omega_n\rho_*^n)$, of [9]:

$$\int_{\rho_*}^{\rho_\delta} |\bar{z}'_\rho|^2(-t')d\rho \leq C_* \int (D_H + D_I)(-t')d\rho = C_* \int P(B_\rho)^2|\psi|d\rho \leq C_*(n\omega_n)^2 K_1\delta =: C_{\text{cent}}\delta. \quad \square$$

6 Conditionality and remarks

Claim: Theorem 1.2 is unconditional. It uses none of: the per- ρ pointwise inputs (1.1), (1.2)=(R), (2.4) of `gmt-inputs-for-metric-closure.md` (open in the no-graph regime); the pointwise quadratic (B^2); Agent 3 (7.1); any selection, graph entry, BDV minimiser, Schauder, or Fuglede expansion.

Structural reason: the centroid kinematic identity (4) has no homothety term, so the bound goes through the unconditional pair (Cauchy–Schwarz on V_ρ ; FMP+FJ+divergence at the centroid).

Remark 6.1 (Sharpness). The pointwise bound (5) is sharp in scaling: for a small translation of a ball, $V_\rho - \bar{V}_\rho$ has dipole structure of size $|\bar{z}'_\rho|$, giving $D_H \sim |\bar{z}'_\rho|^2$.

Remark 6.2 (Outside the smallness regime). If $\delta_T(\Omega) > \delta_0$, then $\mathcal{A}(\Omega) \geq c(n) > 0$ by FMP, and (F) is trivial after enlarging C_{cent} .

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